

Annotation as the Confidence Fulcrum: How AI Systems Classify Digital Content and Why It Determines Recommendation Outcomes

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Abstract

This paper presents an empirical model of how AI systems classify digital content at *annotation*, the processing stage where crawlers label content across multiple dimensions before any ranking, recruitment, or recommendation decision is made. The methodology combines three disciplines: empirical observation (27 years of tracking algorithmic brand representation, measured via publicly viewable instruments and a proprietary dataset; see Section 2.4), analytical reasoning (working backwards from observable downstream needs to necessary annotation inputs), and purpose-built infrastructure (the Kalicube Pro platform, designed to test annotation hypotheses at scale). The paper models *functional roles* within content processing pipelines, not internal implementations of any specific platform. Dialogue with platform engineers at Google and Microsoft provides constraint and sanity checks, not evidentiary confirmation (see Section 1.3).

The paper identifies annotation as the structural fulcrum of the content processing pipeline: the stage where neutral classification creates the competitive scorecard that conditions outcomes at every downstream stage. We introduce several constructs: Annotation-Time Grounding (simultaneous cross-referencing of web index, Knowledge Graph, and parametric model knowledge during crawl-time classification), First-Impression Persistence (the tendency of initial classifications to crystallise and resist subsequent correction), the Annotation Cascade (hierarchical routing through increasingly specialised Small Language Models), the Editorial Grace Window (a brief post-publication period during which classifications remain fluid), and a four-class annotation taxonomy (A1–A4; Section 3.1) that scopes the paper’s claims. The paper also proposes Three-Model Annotation (simultaneous Subject, Entity, and Concept SLM classification) as an architectural model consistent with observed behaviour and, if not already implemented, a recommendation. All claims carry explicit evidential-status labels (Section 1.3). The practical implication is that most optimisation strategies address post-annotation outcomes (ranking, display, click-through) while the confidence that conditions those outcomes is built or destroyed at annotation.

Keywords: annotation, AI recommendation, knowledge graphs, content classification, small language models, entity optimisation, cascading confidence, NEEATT, machine-created ontologies, downstream dependency

1. Introduction

1.1 The Problem: Inconsistency Explained

When Rand Fishkin published research in early 2026 (Fishkin, 2026) demonstrating that AI recommendations are inconsistent across platforms and sessions, the finding formalised a pattern practitioners had been observing but could not explain: the same query, asked of the same system minutes apart, produces different brand recommendations.

The inconsistency is not a bug. It is consistent with a hard-bottleneck confidence architecture where content passes through multiple gates between publication and recommendation. At least three plausible architectures produce this behaviour: multiplicative scoring (where individual factor scores are multiplied, so a near-zero score on any factor effectively eliminates the content), min/AND bottlenecks (where the lowest-scoring dimension caps the overall score), and mixture-of-experts with thresholds (where specialist sub-systems each impose minimum confidence requirements). Gary Illyes (Analyst, Google Search team) described multiplicative combination at SMS Sydney in 2019 (Barnard, 2019a; Illyes, 2019), though whether this applies globally or within specific subsystems is not confirmed. The critical insight is architecture-independent: under any hard-bottleneck model, a single dimension at or near zero produces disproportionately worse outcomes than multiple dimensions at moderate confidence. The predictions in this paper hold under all three architectures.

This paper does not describe the full pipeline architecture (which is the subject of separate work and patent filings). It focuses on a single stage within that pipeline: annotation. The author's research identifies annotation as the structural fulcrum of the entire processing chain, the stage where classification quality conditions outcomes at every subsequent stage, and the stage that has received almost no attention from the optimisation industry.

1.2 What This Paper Contributes

The paper formalises a model of how annotation works at scale, built on a methodology that combines empirical observation, analytical reasoning from downstream needs, and purpose-built testing infrastructure (Section 2). The specific contributions are:

The Downstream Dependency Principle: Annotation dimensions exist because downstream processes need them. New dimensions are identified by reasoning backwards from observable system requirements (Section 4.2).

The Indexing Annotation Hierarchy: A working taxonomy of 40+ dimensions organised into five functional levels, each performing a different operation on content. This is a research artefact, not a claim about any specific engine's internal dimension count. The original 24 were affirmed at a high level by a platform engineer; the expansion to 40+ is the author's analytical contribution.

The Annotation Cascade: Hierarchical routing through specialised Small Language Models (SLMs), with inherited context at each level.

Three-Model Annotation (proposed): Simultaneous classification by Subject, Entity, and Concept SLMs.

First-Impression Persistence: The tendency of initial classifications to crystallise, including the Editorial Grace Window during which reclassification is fluid.

Annotation-Time Grounding: The simultaneous cross-referencing of web index, Knowledge Graph, and SLM parametric knowledge during crawl-time classification.

NEEATT: A proposed decomposition of entity authority into six independently scored dimensions (Notability, Experience, Expertise, Authority, Trustworthiness, Transparency), each evaluated both generally and topic-specifically.

These mechanisms carry different evidential weight, ranging from engineer-affirmed to the author's original proposed architectures (see Section 1.3 and Section 2.5).

1.3 Scope & Epistemic Status

This paper models functional roles within content processing pipelines. It does not claim to describe the internal implementation of any specific platform. The constructs presented here are the author's abstractions, built from sustained observation and analytical reasoning, and validated against observable behaviour across multiple platforms.

Claims throughout this paper carry one of five evidential labels:

Engineer-affirmed at a high level. Affirmed in direct dialogue with platform engineers as directionally correct. Notes are retained; quotations are paraphrased. Affirmation provides constraint and sanity checks, not evidentiary confirmation. Engineers may affirm general direction without endorsing specific architectural claims.

Observation-supported. Consistent with longitudinal data in the Kalicube Pro database and validated through controlled experiments, but not directly affirmed by engineers.

Explanatory hypothesis. The author’s proposed explanation for a specific observed phenomenon, consistent with the model and generating testable predictions that distinguish it from alternative explanations.

Predictive architecture. Derived from the question “how would I build this system given the constraints and observed behaviour?” Observed behaviour is consistent with these models, but alternative implementations could produce the same observed behaviour.

Proposed architecture. The author’s original proposals for how systems could or should be organised. Included because observed behaviour is consistent with them and they constitute recommendations regardless of current implementation.

The detailed evidential status of each specific claim is tabulated in Section 13. Where the paper uses analytical reasoning from downstream needs to infer upstream annotation requirements, this constitutes abductive inference (inference to the best explanation given available evidence), not deductive proof. The inference “a downstream process requires input X, therefore X must be created somewhere upstream; we hypothesise annotation as the most plausible locus under cost constraints” is the governing analytical method.

1.4 Temporal Context

The framework presented here has been developing since 1998, when the author began systematic observation of search engine behaviour. The core insight (that algorithms build confidence through repeated exposure to corroborated information from trusted sources, analogous to how children learn) emerged from the author’s experience creating educational children’s content (Boowa and Kwala, 1998 to 2012), crystallised privately in 2013, and was first articulated publicly in 2017. Specific terminology and concepts formalised in this paper have been developing across the author’s published work since 2012 (Brand SERP, formalised in Barnard, 2022a), 2017 (the author began using the term Answer Engine Optimisation in industry materials; independent third-party coverage documents the term and attributes it to the author by early 2018; Sentance, 2018; and the pedagogical model of algorithmic optimisation), 2023 (the UCD Framework of Understandability, Credibility, and Deliverability, peer-reviewed in Barnard & Artz, 2023), and 2025 (the Indexing Annotation Hierarchy and Annotation Cascade; Knowledge Panel methodology formalised in Barnard, 2025). This paper consolidates the annotation-specific elements of that body of work into a unified model.

2. Methodology: Three Disciplines Applied to Algorithmic Systems

This paper does not claim to reverse-engineer the internal workings of proprietary AI systems. The methodology is empirical modelling: constructing predictive models from sustained observation, validating them through analytical reasoning, and building the infrastructure to test them at scale. The work draws on three complementary disciplines that, taken together,

constitute an integrated approach that combines long-horizon observation, abductive modelling, and large-scale instrumentation.

2.1 Empirical Observation

Twenty-seven years (1998 to present) of daily observation of algorithmic behaviour across search engines, Knowledge Graphs, and Large Language Models. The methodology is Darwinian: sustained, systematic observation across real entities, real content, and real competitive environments, at a scale and duration that allows patterns to emerge that shorter observation windows cannot detect.

Google’s public evolution from keyword matching to semantic understanding, through BERT (Devlin et al., 2019) and MUM (Nayak, 2021), confirms the trajectory toward multimodal, multidimensional content classification that this paper’s annotation model describes. The Kalicube Pro database, collecting data since 2015, tracks brand representation across Google Search, Google AI Mode, ChatGPT, Perplexity, Grok, You.com, Gemini, and Claude. Section 2.4 describes the dataset, its instruments (three of which are publicly viewable), its data sources, definitions, and known biases.

Systematic experimentation complements observation. Controlled tests of how changes to entity representation (structured data modifications, content updates, third-party corroboration campaigns, Knowledge Graph corrections) produce measurable changes in algorithmic behaviour across platforms. The database provides longitudinal tracking that enables before-and-after measurement at a scale unavailable to individual practitioners.

Direct engineer dialogue provides constraint and sanity checks. Sustained conversations with engineers who build and maintain the systems being modelled. At Google: Gary Illyes (Analyst, Google Search team), who described multiplicative confidence combination at SMS Sydney in 2019; and John Mueller (Search Relations). At Microsoft/Bing: Fabrice Canel (Principal Program Manager, responsible for crawling, processing, and indexing; see Canel, 2021; 2025), Nathan Chalmers, Frederic Dubut, Ali Alvi, Ihab Rizk (Microsoft Clarity), and Ragu (Bing Search). Additional observations drawn from public presentations by Martin Splitt (Developer Advocate, Google Search Relations team). Notes are retained; quotations are paraphrased unless otherwise indicated. These conversations provide directional affirmation and, in some cases, identification of dimensions or mechanisms the author had not mapped. Engineer dialogue is treated as *context*, not evidence. The model’s validity rests on observable predictions and falsifiable tests (Section 12). This approach draws on the tradition of abductive reasoning described by C.S. Peirce: inference to the best explanation given available evidence (Peirce, 1903; Lipton, 2004).

2.2 Analytical Reasoning: Working Backwards from Downstream Needs

The methodology's distinguishing analytical characteristic is reasoning backwards from observable outcomes to necessary causes, in the tradition of abductive inference: if a downstream system demonstrably makes a decision, some upstream representation is required; we hypothesise annotation as the most plausible locus under cost constraints.

The core analytical method: if a downstream process demonstrably requires a specific input to make its decision, that input must be created somewhere upstream. If no other processing stage creates it, annotation is the most plausible locus because annotation is the processing stage that examines content directly.

When the author identifies a new annotation dimension, the reasoning is: what does the downstream stage need that it could not produce itself? Ranking needs authority scores. Recommendation needs claim verification. AI-generated responses need quotability assessments. Entity disambiguation needs cultural context. Each of these downstream needs implies an annotation dimension. Reasoning backwards from what the system demonstrably does to what it must therefore classify is abductive inference, not deductive proof, and the resulting hypotheses are subject to the falsification tests described in Section 12.

This analytical method also generates testable predictions. If the model predicts that a particular annotation dimension must exist (because a downstream process needs it), and observed behaviour is consistent with that dimension's presence, confidence in the prediction increases. The question is always: "if I were building a system to solve this problem at this scale with these constraints, how would I architect it, and does the observed behaviour match?" This methodology has produced a documented track record of accurate predictions, including early framing of optimisation for direct-answer interfaces under the label Answer Engine Optimisation; the term is independently documented in industry press by early 2018 (Sentance, 2018), well before generative-answer systems became mainstream.

2.3 Invention: Building the Infrastructure to Test at Scale

Observation reveals patterns. Analysis explains them. But without infrastructure to test predictions systematically across millions of entities, the methodology remains anecdotal. The Kalicube Pro platform is purpose-built infrastructure for testing annotation-related hypotheses at scale. The 16 INPI filings document implementation hypotheses embedded in the platform; they are not presented as evidence, but as formalised system designs derived from the research cycle. The UCD scoring system, the constitutional sandwich prompt architecture, the citation analysis pipeline: each encodes a tested hypothesis into working infrastructure.

This matters methodologically because it closes the loop between observation, analysis, and validation. A prediction derived from analytical reasoning is tested through the infrastructure against millions of entities. The results feed back into observation, refining or rejecting the

hypothesis. This cycle of observe, deduce, build, test, and refine has been running continuously since 2015.

2.4 Data & Measurement

The following instruments underpin the observations in this paper. Three are publicly viewable; reproducibility notes below specify what can and cannot be independently replicated.

Knowledge Graph Sensor (<https://kalicube.pro/tools/knowledge-graph-sensor>). Measures the daily volatility and size of Google’s Knowledge Graph as observed via its public API. Tracks three metrics: magnitude (how many entities saw a confidence score change), amplitude (significance of change per affected entity), and size (estimated order-of-magnitude of identifier-space coverage observable via the Knowledge Graph API response space). The current size estimate is on the order of tens of billions of identifiers; this is a proprietary projection from monitored API responses across a dataset of several thousand known entities, not an official Google figure. Knowledge Graph algorithm updates identified by this sensor have been independently reported in third-party publications (Barnard, 2019b; Barnard, 2023b; Barnard, 2024). The sensor has been operational since 2020 and the data is publicly viewable.

Knowledge Panel Attribute Sources (<https://kalicube.pro/tools/knowledge-panel-attributes>). Tracks the provenance of information displayed in Google Knowledge Panels. As of February 2026: 74% third-party human-curated sources (primarily Wikipedia and Wikidata), 11% Google’s own human-curated sources (Google Business, Google Podcasts, and similar products), and 15% WebFacts (algorithmically deduced by Google’s Knowledge Graph algorithms from trusted source data). The author interprets the WebFacts category as a proxy for the machine-extrapolated growth layer of the Knowledge Graph discussed in Section 7.2; this interpretation is based on observable characteristics but is not confirmed by Google.

Trusted Sources (isReference Domains) (<https://kalicube.pro/tools/trusted-sources>). Identifies which domains Google recognises as entity validation sources. Of 1,738,093 domains monitored, 62,202 qualify as isReference domains (3.65%).

Internal platform data. The Kalicube Pro platform processes data from Google’s Knowledge Graph API, Authoritas (Google and Bing SERPs), SERP API, Wikidata, Wordlift, Common Crawl, and the Kalicube Web Crawler. AI platform data (ChatGPT, Perplexity, Grok, You.com, Claude, Microsoft Copilot, Gemini) is collected via publicly observable outputs. The dataset currently spans 73 million entity profiles and 25 billion data points. The UCD analytical framework underpinning the scoring system has been peer-reviewed (Barnard & Artz, 2023).

Known biases and limitations. The dataset is English-language-dominant and skews toward commercially active entities. Geographic coverage is strongest in North America and Europe. The data is observational: measurements reflect what platforms surface publicly, not internal

classification states. The annotation dimensions proposed in this paper are inferred from these observable proxies, not measured directly.

Reproducibility. The three public instruments are viewable by any third party. Internal platform data is proprietary and not independently reproducible, though the annotation dimensions inferred from it generate falsifiable predictions (Section 12) that can be tested on independent datasets.

Instrument sensitivity. The following conditions would affect the instruments' validity: if Google changes its Knowledge Panel markup or extraction pipeline, the attribute source distribution would shift abruptly (detectable as a discontinuity in the time series). The author monitors for these conditions and has documented several instances of genuine structural changes (Barnard, 2023b; Barnard, 2024).

2.5 Evidential Status of Claims

The evidential-status framework introduced in Section 1.3 is applied throughout. This section provides the detailed evidence for each category.

Engineer-affirmed at a high level. Fabrice Canel (Principal Program Manager, Microsoft Bing) affirmed in conversation: (a) annotation as a distinct processing stage in the content processing pipeline; (b) the principle that annotation classifies without judging ("The bot tags without judging. Filtering happens at query time"); (c) the existence of more than 24 annotation dimensions, with three specific additions (Audience Suitability, Ingestion Fidelity, Freshness Delta); (d) the author's five-level functional organisation as "about right"; and (e) the Editorial Grace Window for URL status changes. Gary Illyes (Analyst, Google Search team) described multiplicative confidence combination at SMS Sydney in 2019; this paper generalises to the broader family of hard-bottleneck architectures. Martin Splitt described the meta description comparison mechanism in a public presentation. Microsoft has publicly described using SLM suites with LLM routing (Yadav, quoted in Microsoft Source, 2024).

Observation-supported. Consistent with longitudinal data and validated through controlled experiments: First-Impression Persistence, Classification Crystallisation (Section 6.3), the author misattribution case study (Section 6.4), and the multi-layer entity identification system.

Explanatory hypothesis. The AI Overviews annotation-readiness explanation (Section 4.2).

Predictive architecture. The Annotation Cascade hierarchy; Annotation Cost Threshold; SLM routing triggers; Index Owner vs. Index Renter dynamic.

Proposed architecture. Three-Model Annotation; Annotation-Time Grounding; Topical Tier 1; NEEATT decomposition; multi-topic dominance weighting.

Non-denial in engineer dialogue (Appendix B) is included for transparency and is not treated as evidence.

2.6 Terminology and Scope

Three terms in this paper carry specific operational meanings. “Knowledge Graph” refers to the entity graph accessible via Google’s public Knowledge Graph API and observable through Knowledge Panels, SERP features, and entity-linked results. “Annotation” refers to the processing stage where crawlers assign multi-dimensional labels to content chunks before any ranking or recommendation decision; the exact boundary between indexing and annotation may vary across implementations, but the structural claim that labelling precedes ranking is consistent with engineer affirmations and observable behaviour. “Confidence” refers to the model’s construct of per-dimension certainty scores that condition downstream outcomes.

3. What Annotation Is (and What It Is Not)

Operational Definition. Throughout this paper, “annotation” refers to the following construct:

Inputs: Crawl-time content (text, structure, metadata) plus contextual signals (entity associations, link graph, Knowledge Graph presence, SLM parametric knowledge).

Outputs: Dimension labels with associated confidence scores, assigned to retrievable chunks and pages. These labels persist in the index independently of any specific query.

Excludes: Ranking decisions, query-time retrieval, user-specific personalisation, and display formatting. These are downstream consumers of annotation outputs, not part of annotation itself.

Observable proxies: Snippet stability across queries, AI citation consistency across sessions, entity resolution accuracy in Knowledge Panels, classification persistence after content updates, and time-to-first-citation for new content.

3.1 Annotation Scope: A Taxonomy

To avoid conflating distinct annotation functions, we distinguish four classes:

A1 — Topical classification and routing. Assigning content to topic domains and routing to specialist models. This is the annotation function that determines which competition pool the content enters.

A2 — Entity linking and resolution. Identifying entities mentioned in content, disambiguating them against the Knowledge Graph, and establishing entity-content associations.

A3 — Claim extraction and relation extraction. Identifying factual assertions, attributing claims to entities, and mapping relationships between entities and concepts.

A4 — Quality and deployment features. Scoring content on dimensions such as quotability, standalone score, verifiability, and freshness — features that determine how content is deployed at query time.

This paper focuses primarily on **A1–A2 and the confidence meta-layer** that spans all four classes. A3–A4 are treated as downstream consumers of A1–A2 outputs: claim extraction (A3) requires entity resolution (A2) to attribute claims, and deployment features (A4) require topical classification (A1) to determine competition pool. The annotation dimensions mapped in Section 4 span all four classes but the architectural claims (Sections 5–7) address the routing and entity-linking layers specifically.

3.2 Annotation as a Distinct Processing Stage

Annotation is not indexing and it is not ranking. Indexing stores content and segments it into retrievable units (“chunks”); annotation assigns multi-dimensional labels to those chunks. Ranking, recruitment, and recommendation happen later, using the labels that annotation has already applied. The distinction matters because annotation is a neutral labeller. It classifies content across dimensions, including quality dimensions, without making selection decisions. Fabrice Canel affirmed the governing principle: “The bot tags without judging. Filtering happens at query time.”

3.3 Annotation Operates on Chunks in Context

At the indexing stage, content is segmented into retrievable chunks. Annotation classifies each of those chunks in the context of the system’s understanding of the page as a whole. The page-level topic, entity associations, and intent provide the frame within which each individual chunk is classified. If page-level understanding is confused (unclear topic, ambiguous entity, mixed intent), every chunk annotation inherits that confusion.

Martin Splitt has described a related mechanism in public presentations: Google uses the page’s meta description as a comparison summary. The system generates its own two-sentence summary of the page using the language model, then compares that against the author-provided meta description. When they match, the system’s confidence in its page-level understanding increases, and that increased confidence cascades into marginally better annotation scores for every chunk on the page. One of thousands of small signals that cumulate.

4. The Indexing Annotation Hierarchy: A Working Taxonomy

4.1 Mapping Methodology

The author has mapped 40+ annotation dimensions organised into five functional categories. *This mapping is a working taxonomy — a research artefact — not a claim about any specific*

engine's internal dimension count or organisation. The original 24-dimension mapping was affirmed as directionally correct by Fabrice Canel, who identified additional dimensions beyond that initial set. Canel also affirmed the five-level functional organisation as “about right,” which the author interprets as directional affirmation that the levels capture real structural distinctions, even if the internal implementation differs in detail.

The expanded dimensions beyond the original 24 are the author’s analytical contributions, derived from the abductive method formalised in Section 2.2: reasoning backwards from downstream needs to necessary annotation inputs. The mapping will be expanded over time as the author identifies additional downstream dependencies.

4.2 The Downstream Dependency Principle

Annotation dimensions do not exist in isolation. Every dimension exists because some downstream process consumes it. This dependency relationship is the key to predicting which dimensions the annotation layer contains: if a downstream stage needs a particular classification to make its decision, that classification must be created at annotation, because annotation is the processing stage that examines the content directly. This is abductive inference: the downstream process demonstrably makes a decision; some upstream representation is required; we hypothesise annotation as the most plausible locus under cost constraints.

This principle operates across three temporal horizons:

Legacy dimensions. Some annotation dimensions may persist from historical downstream needs that no longer exist in their original form. These legacy dimensions are computationally cheap to maintain and may serve as redundant signals or fallback classifiers.

Current dimensions. These serve active downstream needs: ranking, recruitment into AI-generated responses, Knowledge Graph population, entity disambiguation. The majority of mapped dimensions fall into this category.

Anticipatory dimensions. Systems may annotate dimensions for projected future downstream needs, building the classification corpus in advance. Retrofitting annotation across billions of already-processed pages is vastly more expensive than adding a dimension to ongoing classification.

This temporal framework offers a structural explanation for one of the most visible failures in recent AI history. Google’s AI Overviews launch in 2024 produced well-documented errors including the notorious recommendation to add glue to pizza. The standard explanation was model quality: the LLM hallucinated. The annotation model suggests a different and more structural explanation: the annotation layer was not ready.

The annotation-readiness hypothesis predicts systematic failure patterns that the model-hallucination explanation does not. Under the hallucination model, errors should be randomly distributed across content types. Under the annotation-readiness model, errors should cluster: (a) failures concentrated in queries requiring source-claim attribution (A3 dimension, which keyword-era annotation never needed); (b) failures concentrated where entity authority decomposition matters (medical, legal, financial); (c) failures disproportionately affecting content from sources with weak Knowledge Graph presence; and (d) progressive improvement over time as annotation dimensions mature, independent of model version updates. The measured rollout of Google's AI Mode through 2025 and 2026, with phased expansion to new query categories, is compatible with the annotation-readiness hypothesis. An alternative explanation is that the phased rollout reflects model capability limitations or legal caution rather than annotation maturity; distinguishing between these requires the failure-pattern analysis described in Section 12.

4.3 The Five Functional Levels

Level 1: Gatekeepers (eliminate). Binary pass/fail dimensions. Examples: temporal scope, language detection, entity resolution, content format (text/video/image/audio), ingestion fidelity (*affirmed by Canel*: whether the bot could parse the content cleanly), safety and policy compliance. Failure at any gatekeeper eliminates the chunk from that query's candidate pool instantly. These are pre-competitive: the content does not enter the race at all. Other gatekeeper dimensions include geographic scope (operating at country → region → city → neighbourhood granularity) and cultural classification (conditioning what content is appropriate and resonant, not merely relevant). Cultural classification is where machine-created topic ontologies become essential, because culture resists human-curated taxonomy and evolves faster than any editorial team could maintain.

Level 2: Core Identity (define). Continuous-score dimensions. Examples: topic classification with dominance weighting (a page is rarely single-topic; the system likely assigns multiple topics with proportional weighting, e.g., 60% marketing, 30% technology, 10% business), entity identification, attributes, relationships, sentiment, claim density. These define what the content means: who is discussed, what facts are stated, how entities relate, what the tone is. Other Level 2 dimensions omitted for brevity.

Level 3: Selection Filters (route). Continuous-score dimensions determining competition pool. Examples: intent category, expertise level, audience suitability (*affirmed by Canel*: routing content to the appropriate audience pool), content depth (overview vs. comprehensive vs. reference). Wrong pool placement means competing against better-matched content. Other Level 3 dimensions omitted for brevity.

Level 4: Confidence Multipliers (rank). Continuous-score dimensions affecting competitive position. Examples: verifiability, provenance, corroboration count, freshness delta (*affirmed by*

Canel: how current the content is relative to the topic's rate of change), citation density. Entity authority is proposed as decomposed into six NEEATT dimensions (Notability, Experience, Expertise, Authority, Trustworthiness, Transparency), each scored both generally and topic-specifically. The general score affects entity recognition; the topic-specific score affects whether the entity's content is treated as authoritative within the classified topic. Other Level 4 dimensions omitted for brevity.

Level 5: Extraction Quality (deploy). Dimensions determining deployment format (A4 class). Examples: sufficiency, standalone score, quotability (whether a chunk can be extracted and stand alone as a citation in an AI-generated response), entity salience, entity role. Other Level 5 dimensions omitted for brevity.

4.4 Confidence as Meta-Dimension

A Confidence Score is attached to every individual annotation at every level. This is the meta-dimension: not what the system thinks the content means, but how certain it is about each classification. Clarity drives confidence. Ambiguity degrades it. A chunk can be correctly classified across all dimensions but still receive low confidence scores if the classifications were difficult to make.

4.5 Beyond 24

The original mapping identified 24 dimensions. Fabrice Canel, when presented with that mapping, responded: "Oh, there is definitely more." He subsequently identified three additional dimensions not in the original mapping: Audience Suitability (Level 3), Ingestion Fidelity (Level 1), and Freshness Delta (Level 4). The author estimates that the actual classification space encompasses hundreds of individual dimensions. The 40+ working taxonomy presented here is substantially more complete than the original 24, but remains a subset.

5. SLM Routing and the Annotation Cascade

5.1 Specialist Models, Not General Models

AI systems do not use a single large language model to classify all content. They route content to specialised Small Language Models (SLMs): domain-specific models that are computationally cheaper, faster, and paradoxically more accurate than general LLMs for niche content. Microsoft's Phi family of SLMs has demonstrated that models with as few as 3.8 billion parameters can outperform much larger models on domain-specific tasks when trained on high-quality, curated data (Abdin et al., 2024). Microsoft has publicly described using "suites of models, where large language models play the role of router, to direct certain queries that require less computing power to small language models" (Yadav, quoted in Microsoft Source, 2024).

Generalist fallback correlates with lower confidence and higher variance in the Kalicube Pro observations. When the system cannot route to a specialist, it defaults to a general model, and generalist annotation consistently produces lower confidence scores across downstream proxies.

5.2 The Annotation Cascade

The specific routing hierarchy described here is the author's predictive architecture of how SLM routing is most likely organised, based on observed behaviour and computational logic. The proposed cascade follows: Site → Category → Page → Chunk. Each level inherits or reroutes the model context from the level above. The crawler does not arrive at a page cold. It arrives with inherited context: what kind of site is this? What section? What topic? Each level narrows SLM selection and either confirms or overrides the routing from above.

Misrouting at any level (assigning the wrong specialist model) produces confident misannotation, which is worse than falling to a general model. A medical SLM applied to financial content will classify with high confidence and be wrong. A general model would classify with lower confidence but at least not produce false authority signals.

5.3 Three-Model Annotation (Proposed Architecture)

The following is the author's proposed architecture. No engineer has described this specific structure, and it may be entirely novel. It is included because observed behaviour is consistent with it and it represents how the author would solve the classification problem given known constraints. Three types of SLM operating simultaneously for each topic:

Subject SLM. Classifies content by subject matter. What is this about?

Entity SLM. Resolves entities and assesses centrality and authority. Who are the key players?

Concept SLM. Maps claims to established concepts and evaluates novelty. What ideas define this space?

When all three return high confidence on the same entity for the same content, annotation cost is minimal. When they disagree, confidence drops and the system falls back to a more general model.

5.4 How to Trigger SLM Routing

If model routing operates as described, the practical implications follow logically. Content that is category-clear early (the author estimates within the first 100 words), uses standard industry terminology, follows structural conventions, and references well-known entities would trigger specialist SLM routing. Content that is topically ambiguous, terminologically creative, or structurally unconventional gets the generalist. This has a counterintuitive practical implication:

novel terminology and creative formatting, which practitioners often assume signal expertise, may actually reduce annotation confidence by preventing specialist model routing.

5.5 Annotation Cost Threshold

The author proposes that when sufficient data density accumulates in a topic, it becomes computationally cheaper for the system to build or cache a specialist SLM than to process through a general model. Microsoft Research demonstrated the underlying principle with the Phi model family: SLMs trained on curated data outperform much larger models trained on undifferentiated web content (Gunasekar et al., 2023). This would explain why niche topics can sometimes achieve higher annotation confidence than broader topics: the specialist model is more focused, and the reference corpus against which it evaluates content is more concentrated.

6. First-Impression Persistence and the Editorial Grace Window

6.1 Classification Crystallisation

When the bot first crawls a page, it selects an SLM, runs its annotation process, assigns confidence scores across all dimensions, and saves that classification. On subsequent crawls, it starts with the previously assigned SLM and previously assigned annotations. The initial classification becomes the baseline against which all subsequent signals are measured. The system does not re-evaluate from scratch on each crawl. It checks consistency with the existing classification, reinforcing it if consistent, requiring sustained and unambiguous counter-signals if not. The author terms this phenomenon First-Impression Persistence.

First-Impression Persistence has been observed across multiple dimensions in the Kalicube Pro database: SLM classification, entity association, topical assignment, and author attribution. The mechanism is computationally logical: re-evaluating every classification from scratch on every crawl would be prohibitively expensive.

6.2 The Editorial Grace Window

Fabrice Canel affirmed that when a URL returns a 404 or is redirected with a 301, the system provides a grace period (observed as approximately one week) during which it assumes the change may be an error that will revert. Canel has also described the scale of the processing challenge: Bing processes over 70 billion URLs per day (Canel, 2023). After the grace period, the new state becomes the persistent baseline.

The author proposes, based on observed behaviour, that the same principle applies to content classification: a brief post-publication window (likely one to several weeks) during which annotation is fluid and receptive to correction. After this window closes, the classification crystallises.

6.3 Noise as the Enemy of Change

Classification can be changed after crystallisation, but thoroughness is the prerequisite. The author conducted a controlled correction in 2025: the phrase “Web Index” was changed to “Search Engine” across approximately ten published articles. Every reference was updated. Zero traces of the old terminology were left behind. Within one month, AI systems were consistently using “Search Engine” in place of “Web Index.” The principle: any residual contradictory signal maintains inertia proportionally. Thoroughness, not time, is the variable that determines whether reclassification succeeds.

6.4 Case Study: Author Misattribution on Search Engine Land

Google briefly attributed several Search Engine Land articles written by Barry Schwartz to the author of this paper. The misattribution appeared in the author’s Knowledge Panel for approximately two days. The mechanism: the author’s name appeared prominently below the articles (author bio placement). The annotation system used a cheap heuristic (first recognised entity name below content equals author) rather than the more expensive author-vector analysis. The grace window allowed correction within two days, suggesting grace period duration is dimension-specific.

7. Annotation-Time Grounding: A Proposed Architecture

7.1 Three-Source Cross-Referencing During Classification

The individual components of this model are established: the web index exists, the Knowledge Graph exists, and SLMs have parametric knowledge. The author’s contribution is the proposal that these three sources are cross-referenced simultaneously during crawl-time classification, not sequentially or at different processing stages. The model rests on the observable predictions and tests described in Section 12.

During crawl-time classification, the author proposes that the bot cross-references against at least three sources:

The web index. The bot would carry prioritised access to the web index during crawling, checking content against existing index knowledge: who links to the page, what context those links provide, how the page’s claims relate to claims on other pages.

The Knowledge Graph. The bot checks annotated entities against the Knowledge Graph during classification. Google introduced the Knowledge Graph in 2012 (Singhal, 2012). By 2020, the graph contained over 500 billion facts across 5 billion entities (Nayak, 2020). If an entity is already in the graph with high confidence, annotation would inherit that confidence. If absent, annotation would start from a substantially lower baseline.

The SLM’s parametric knowledge. Claims that align with the SLM’s existing knowledge receive higher confidence annotations. Claims that contradict it are flagged. Novel claims receive lower confidence until corroboration accumulates.

This triple cross-referencing, here termed Annotation-Time Grounding, would mean that annotation quality is not solely a function of how well content is written. It would be a function of how well the publishing entity is already represented across all three sources.

7.2 Entity Identification Layers in the Knowledge Graph

The Knowledge Graph is not a monolithic system. Entity identification operates across multiple layers with different provenance and confidence levels:

/m/ identifiers: human-curated. The original entity layer, manually created and maintained. These carry the highest baseline confidence.

/g/ identifiers: hybrid provenance. Some seeded from curated databases (Wikipedia, Wikidata, MusicBrainz); some from Google’s own products; and some machine-generated: the system extrapolates new entities when it encounters sufficient corroborated evidence. The Kalicube Pro Knowledge Panel Attribute Sources tracker shows 15% of attributes sourced from WebFacts (algorithmically deduced), which the author interprets as a proxy for this machine-extrapolated /g/ layer.

Topic identifiers: the /g/ wrapper hypothesis. The author’s working hypothesis is that /g/ identifiers serve as a public-facing synonym for internal topic classifications (/t/ identifiers). The “Jason Barnard” topic carries a /g/ identifier (/g/11hf5y7764) that returns nothing from the Knowledge Graph API, supporting this interpretation. The relationship between topic ontologies, disambiguation clusters, and SLM routing represents a promising research direction explored further in Appendix A.

The annotation implications are significant. When the bot resolves an entity during classification (Level 1 Gatekeeper: entity resolution), it is navigating a multi-layered identification system where the provenance of the entity’s Knowledge Graph entry directly affects the confidence inherited by the annotation. This layered confidence inheritance is consistent with observed behaviour: well-established entities consistently receive higher downstream performance on new content than emerging entities.

7.3 The Flywheel

If Annotation-Time Grounding operates as described, the feedback loop is self-reinforcing. Strong presence across the three grounding sources leads to higher annotation confidence on new content, which leads to better recruitment downstream, which strengthens presence across the sources. This compounding advantage is consistent with observed behaviour in the Kalicube Pro

database. The practical corollary: Knowledge Graph optimisation, web authority building, and content that trains SLMs are not three separate strategies. They are three inputs to the same annotation-time grounding process.

7.4 Topical Tier 1 and the Crawl Companion Index

If the bot carries prioritised web index access during crawling, that subset would constitute a Tier 1 index: the most authoritative portion of the full index for a given topic. The author proposes that this Tier 1 subset is topic-specific, paralleling SLM specialisation. The SLM knows what claims are standard; the Tier 1 subset knows which sources are authoritative. Together, they would form the annotation toolkit for that topic.

7.5 Knowledge Graphs Are Not Exclusive to Google

The Knowledge Graph, LLM, and Search Engine trinity (the Algorithmic Trinity) is an architectural pattern, not a Google-specific phenomenon. In November 2025, Andrea Volpini (CEO, WordLift) intercepted ChatGPT's internal Server-Sent Event data streams and documented what appears to be an operational entity layer running beneath every conversation (Volpini, 2025). Academic research on knowledge-grounded language models consistently demonstrates that graph-structured knowledge reduces hallucination and improves factual accuracy (Peng et al., 2024; Edge et al., 2024). Annotation-Time Grounding is therefore not a Google-specific phenomenon.

8. Platform-Specific Annotation Behaviour

8.1 Index Owners vs. Index Renters

Google and Bing own their crawling infrastructure, indexes, and Knowledge Graphs. They can afford grace periods, scheduled re-checks, and maintained temporal state. OpenAI, Perplexity, and engines that rent index access operate on a different model. The author proposes this creates a two-speed dynamic:

Slow Boolean gate: Whether content is in the candidate pool at all depends on the index those engines draw from. They inherit the index owner's annotation decisions, including classification inertia.

Fast display layer: What these engines show when they cite content is fetched in real time at query-time grounding. Displayed content is fluid even when Boolean presence in the candidate pool is slow to change.

The distinction means that optimising annotation for Google and Bing simultaneously optimises annotation for every engine that draws from their indexes.

9. The Annotation Gap: What the Industry Optimises vs. What Conditions Outcomes

The SEO industry’s dominant optimisation strategies address what happens after annotation: ranking signals, display optimisation, click-through rate, and conversion. These are important. They are also downstream. They depend entirely on the quality of annotation that has already been applied at crawl time. The shift from keyword-matching to entity-based understanding, described by Metzler et al. (2021) as the transition from “dilettante” search to “expert” information retrieval, demands a corresponding shift in optimisation strategy toward the classification stage where that understanding is built.

In the broader content processing pipeline, annotation sits at the structural midpoint. Before annotation, processing stages are infrastructure (can the system find, access, render, and store the content?). At annotation, the system classifies content across multiple dimensions, and those classifications create the competitive scorecard. Post-annotation, every processing stage uses those labels to make competitive decisions. Misclassification at annotation propagates through every downstream stage.

10. Prior Art and Temporal Evidence

The concepts formalised in this paper have been developing across the author’s published work over more than a decade: 2012 (Brand SERP; formalised in Barnard, 2022a), 2013 (pedagogical model of algorithmic optimisation), 2017 (Answer Engine Optimisation; independently documented in Sentance, 2018), 2019 (multiplicative confidence model; Barnard, 2019a), 2023 (UCD framework; peer-reviewed in Barnard & Artz, 2023), 2025 (Indexing Annotation Hierarchy, Annotation Cascade, NEEATT; Knowledge Panel methodology in Barnard, 2025), and 2026 (Downstream Dependency Principle, Annotation-Time Grounding, First-Impression Persistence, and remaining constructs formalised in this paper).

11. Limitations

This model is empirical, not verified against source code. The author does not have access to the internal architectures of Google, Microsoft, OpenAI, or other AI systems. The model’s validity rests on its predictive accuracy, explicit engineer affirmations, consistency with observable behaviour, and the falsifiable tests described in Section 12. As detailed in Sections 1.3 and 2.5, claims carry different evidential weight: some are engineer-affirmed at a high level, some are observation-supported, some are predictive architecture, and some are the author’s original proposed architectures that may represent novel contributions rather than descriptions of existing systems.

The 40+ annotation dimensions are the author's working taxonomy, expanded from an original 24 that were affirmed as directionally correct. The expanded dimensions (including NEEATT and multi-topic weighting) are the author's proposed additions. The Editorial Grace Window durations cited are observed estimates, not confirmed parameters. The role of machine-created ontologies in annotation classification represents a substantial area for future investigation (see Appendix A).

12. Falsifiable Predictions and Planned Tests

The model described in this paper generates specific, testable predictions. Each test is designed to be reproducible by any researcher with access to multi-platform brand tracking and longitudinal SERP/AI citation data.

12.1 Editorial Grace Window Test

Prediction: Content corrections made within the first week after publication produce faster and more complete reclassification than identical corrections made after 30 days.

Protocol: Publish near-identical pages with controlled factual errors. Apply corrections at day 2, day 10, and day 30. Measure reclassification speed, completeness, and residual persistence across Google, Bing, ChatGPT, Perplexity, and Gemini.

Falsification: If corrections at day 30 produce reclassification at comparable speed and completeness to corrections at day 2, the Grace Window hypothesis is weakened.

12.2 First-Impression Persistence Test

Prediction: Initial content with deliberate topical ambiguity produces a classification that persists even after the ambiguity is removed.

Protocol: Publish paired pages: one with clear topical signals, one with deliberate ambiguity. After initial indexing, update the ambiguous page to match the clear page exactly. Measure convergence time.

Falsification: If convergence occurs within one crawl cycle, First-Impression Persistence is not operating as described.

Preliminary status: Two observations consistent with this prediction are documented in Sections 6.3 and 6.4.

12.3 SLM Routing Test

Prediction: Content using standard industry terminology receives higher annotation confidence (measured via downstream proxies) than semantically equivalent content using novel or creative terminology.

Protocol: Publish matched pairs with identical claims but different terminology (“search engine optimisation” vs. “algorithmic visibility enhancement”). Measure downstream confidence proxies over 90 days.

Falsification: If creative terminology produces equivalent or higher downstream confidence proxies, the SLM routing trigger hypothesis is weakened.

12.4 Annotation-Time Grounding Test

Prediction: New content published by an entity with strong Knowledge Graph presence receives higher annotation confidence than identical content from an entity with weak or absent KG presence.

Protocol: Identify matched entity pairs: similar domain authority, similar content quality, but divergent KG presence. Publish comparable new content simultaneously. Measure AI citation frequency and recommendation consistency over 60 days.

Falsification: If KG presence shows no correlation with AI citation frequency or recommendation consistency, the hypothesis is weakened.

Data availability: The Kalicube Pro database contains matched-pair data. Preliminary results expected 2026.

12.5 Topic-Cluster Dominance Test

Prediction: Entities that dominate their name cluster receive correct entity resolution more frequently in AI citations than entities sharing a contested name cluster.

Protocol: Compare cohorts: entities with >80% Brand SERP dominance vs. entities with 50/50 splits. Measure entity resolution accuracy and misattribution rate across platforms.

Falsification: If name-cluster dominance shows no correlation with entity resolution accuracy, the disambiguation cluster model is weakened.

12.6 Consolidation as Annotation Noise Reduction Test

Prediction: Consolidating similar pages produces annotation confidence improvements that exceed what “reduced cannibalisation” alone would predict.

Protocol: Consolidate and measure not only ranking recovery but also AI citation frequency, recommendation consistency, and Knowledge Panel accuracy.

Falsification: If consolidation improves rankings but shows no improvement in AI citation frequency or recommendation consistency, the annotation noise reduction explanation adds nothing.

12.7 Hard-Bottleneck Confidence Test

Prediction: Entities with a single critical annotation failure show disproportionately worse downstream outcomes than entities with multiple minor annotation weaknesses.

Operational definitions: “Single critical failure” = one dimension below threshold τ , proxied by: entity resolution error (entity not recognised or misidentified in AI citations), safety flag (content excluded from recommendation pools), or topical misclassification (content appearing in wrong competition pool, observable via SERP feature mismatch). “Distributed weakness” = $k \geq 3$ dimensions in mid-range confidence, proxied by: moderate AI citation frequency, inconsistent but non-zero recommendation visibility, and partial entity resolution accuracy.

Protocol: Categorise entities by failure pattern. Compare downstream outcomes (SERP visibility, AI citation rate, recommendation frequency) relative to overall annotation quality.

Falsification: If “single critical failure” and “distributed weakness” entities show proportionally similar downstream outcomes, the hard-bottleneck model is weakened.

12.8 Research Timeline

Tests 12.4 and 12.5 can be executed as correlation analyses on the existing Kalicube Pro dataset; preliminary results expected 2026. Tests 12.1, 12.3, 12.6, and 12.7 require controlled experiments; the author intends to execute these beginning in 2026, with results expected by late 2026 or early 2027. The author invites collaboration from academic researchers with access to independent datasets.

Table 2: Mechanism–Proxy–Direction–Test Mapping

Mechanism	Observable Proxy	Expected Direction	Planned Test
Annotation Cascade (SLM routing)	Downstream confidence proxies (SERP stability, AI citation rate)	Standard terminology → higher confidence than creative synonyms	12.3
First-Impression Persistence	Classification convergence time after correction	Initially ambiguous pages converge slower than initially clear pages	12.2

Editorial Grace Window	Reclassification speed after correction	Day-2 corrections faster than day-30 corrections	12.1
Annotation-Time Grounding	AI citation frequency for new content	Strong KG presence → higher citation rate	12.4
Topic-cluster dominance	Entity resolution accuracy in AI citations	High name-cluster dominance → lower misattribution rate	12.5
Consolidation as noise reduction	AI citation + recommendation consistency post-consolidation	Improvements exceed ranking-only recovery	12.6
Hard-bottleneck confidence	Downstream outcomes by failure pattern	Single critical failure → disproportionately worse than distributed weakness	12.7
Annotation-readiness (AI Overviews)	Error distribution across content types	Errors cluster by annotation-dimension maturity, not randomly	Failure-pattern analysis

13. Evidence Summary

Claim	Evidence Type	Source	Planned Test
Annotation is a distinct processing stage	Engineer-affirmed	Canel (explicit)	N/A (affirmed)
"The bot tags without judging"	Engineer-affirmed	Canel (explicit)	N/A (affirmed)
24+ annotation dimensions exist	Engineer-affirmed	Canel (affirmed original 24, added 3)	Ongoing mapping
Five-level functional organisation	Engineer-affirmed	Canel ("about right")	N/A (directional)
Editorial Grace Window (URL status)	Engineer-affirmed	Canel (explicit)	Test 12.1
Hard-bottleneck confidence scoring	Engineer-described	Illyes, SMS Sydney 2019	Test 12.7
SLM suites with LLM routing	Public statement	Microsoft (Yadav, 2024)	Test 12.3
First-Impression Persistence	Observation-supported	Kalicube Pro data	Test 12.2
Classification Crystallisation	Observation-supported	Controlled correction (6.3)	Test 12.2

AI Overviews annotation-readiness	Explanatory hypothesis	Author’s analysis	Failure-pattern analysis
Annotation Cascade	Predictive architecture	Author’s model	Test 12.3 (indirect)
Three-Model Annotation	Proposed architecture	Author’s proposal	Indirect via proxies
Annotation-Time Grounding	Proposed architecture	Author’s proposal	Test 12.4
NEEATT decomposition	Proposed architecture	Author’s proposal	Component analysis
Topic-cluster disambiguation	Proposed architecture	Google Trends observable	Test 12.5

14. Conclusion

Annotation is the confidence fulcrum. It is where AI systems decide what content means, which entity it belongs to, what claims it makes, and how confident those classifications are. Every subsequent processing stage depends on annotation quality. Optimising downstream without securing annotation is structurally equivalent to competing with an incomplete scorecard.

The Downstream Dependency Principle is the paper’s governing insight: annotation dimensions exist because downstream processes need them. This principle is both diagnostic (explaining why certain dimensions exist) and predictive (identifying dimensions that must exist but have not yet been mapped). As AI systems add new downstream capabilities (conversational recommendation, agentic task completion, multimodal synthesis), new annotation dimensions will be required. The annotation layer is not merely a reflection of current system capabilities. It is a leading indicator of future ones.

The mechanisms introduced in this paper, spanning engineer-affirmed processes, observation-supported patterns, explanatory hypotheses, predictive models, and proposed architectures, provide a vocabulary and framework for addressing what the author believes is the most consequential and least examined processing stage in AI-era content optimisation.

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Appendix A: Name-Cluster Dynamics and Topic Identifiers

This appendix provides the extended example and analysis supporting the topic-cluster disambiguation model described in Section 7.2.

The “Jason Barnard” topic is the conceptual cluster of conversation around the name string. This cluster is the weighted composite of every individual who shares the name and all of their associated topic spaces. Each name-sharer contributes facets to the cluster. The author contributes the dominant facets: Brand SERPs, Knowledge Panels, Knowledge Graphs, AI brand management. A UK-based music podcaster contributes a genuinely present music facet. An academic and an ice hockey player contribute relatively little topical weight. A lithium energy CEO, whose company is listed on the Canadian stock exchange and who appears increasingly in financial news, contributes a growing facet that overlaps with the author’s on one dimension: both are CEOs.

The “CEO” facet is shared territory within the cluster. A query about “Jason Barnard CEO” triggers a disambiguation competition between two entities that both legitimately contribute to that facet. The author’s CEO facet is grounded in digital brand intelligence; the lithium energy CEO’s is grounded in mining and Canadian capital markets. “Jason Barnard CEO Kalicube” resolves cleanly. “Jason Barnard CEO” without further context is a genuine disambiguation challenge.

The collective topic signature, as of this writing, is: Brand SERPs, Knowledge Panels, Knowledge Graphs, AI brand management, music, and CEO. That signature is not static. Every entity’s weight is relative, not absolute. Dominance is maintained through ongoing activity, not established permanently.

Every name string creates a unique cluster. A common name like “John Smith” produces an impossibly diffuse cluster where disambiguation is computationally expensive on every encounter. A rare name produces a concentrated cluster. This is where machine-created ontologies outperform human-curated taxonomy: the system can discover name-cluster patterns from observed data at a scale no human editorial team could manage.

The annotation implication: topic-cluster dominance, built through sustained niche authority, directly affects annotation confidence at the entity resolution stage. An entity that dominates its name cluster receives correct disambiguation by default; an entity that shares a contested cluster must work harder to trigger correct resolution.

Appendix B: Non-Denial in Engineer Dialogue

When presented with the specific hypothesis that the system queries the Knowledge Graph at annotation time, Fabrice Canel responded: “No official comment.” This is included for transparency and to document that the hypothesis was posed and not refuted. It is not treated as

evidence. Canel's response may reflect corporate policy, NDA constraints, or PR guidance rather than implicit confirmation. The Annotation-Time Grounding model (Section 7) rests on observable predictions and the falsifiable tests described in Section 12, not on this non-denial.

Author

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